

| Parameter         | Value                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------|
| Forward model     | multislice through vitreous ice                                                                 |
| Transfer function | weak-phase CTF, CTFFIND/RELION convention                                                       |
| chi(k)            | $\pi \cdot \lambda m \cdot d f \cdot k^2 - 0.5 \cdot \pi \cdot C s \cdot \lambda m^3 \cdot k^4$ |
| Envelopes         | temporal, spatial, B-factor, dose damage                                                        |
| Microscopes       | crispr, cryos-cfeg, glacios, talos-arctica, talos-l120c, cs-corrected                           |
| Detectors         | K3, K2, Falcon4, Falcon4i, Falcon3EC, Apollo, DE64, Ceta, ideal                                 |
| Parameters        | detector-specific DQE(q) and MTF(q); realistic shapes, not vendor curves                        |
| Slice thickness   | 40 Å                                                                                            |
| Inelastic loss    | thickness-dependent                                                                             |
| Validation        | recovered from the simulated power spectrum, R2 0.986, MAE 0.163 um                             |